



**A HYBRID RULE-BASED AND MACHINE LEARNING
FRAMEWORK FOR DETECTING HUMAN-LAYER
CYBERSECURITY POLICY VIOLATIONS IN MILITARY-
STYLE NETWORKS**

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17 December 2025

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Abstract

Most Militaries around the world are increasingly relying on complex information systems to support command, control, and operational decision-making. While significant effort has been invested in securing these systems through technical controls, many cyber incidents continue to arise from human-layer cybersecurity policy violations, such as credential misuse, inappropriate privilege usage, and unsafe interactions with information systems. Existing approaches for managing these risks rely largely on awareness training and post-incident investigation, offering limited capability for continuous detection and measurable enforcement of cybersecurity policy compliance.

This study proposes the design and implementation a hybrid framework that combines rule-based policy enforcement with machine learning-based risk scoring to detect human-layer cybersecurity policy violations in military-style networks. The framework treats human behaviour as a measurable cybersecurity control surface by mapping clearly defined policy requirements to observable system events, including authentication activity, privilege use, and endpoint interactions such as USB insertions, admin privilege use. Rule-based mechanisms are used to identify explicit policy violations, while a modest and explainable machine learning component supports the identification and prioritisation of anomalous or high-risk behavioural patterns.

The research adopts a design-science approach and demonstrates the feasibility of the proposed framework through a constrained prototype evaluated using a laboratory testbed and realistic synthetic data. Evaluation focuses on detection accuracy, false-positive rates, and the framework's ability to support timely and auditable cybersecurity governance decisions without reliance on classified systems or data.

The study contributes to cybersecurity research by showing how hybrid detection techniques can be applied to the human layer in a controlled and explainable manner. Practically, it provides military organisations with a structured and measurable approach for strengthening human-layer cybersecurity controls while respecting the governance, accountability, and assurance requirements of military-style environments.

Keywords: *Human-Layer Cybersecurity, Cybersecurity Policy Violation, Hybrid Detection Framework, Hybrid Rule-Based, Machine Learning Risk Scoring, Military-Style Networks*

CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

The contemporary military environment is increasingly shaped by digital transformation, network-centric operations, and the formal recognition of cyberspace as an operational domain alongside land, sea, air, and space. Modern armed forces rely extensively on information and communication technologies to support command and control, intelligence collection, logistics coordination, and operational decision-making (Healey, 2013; Johnson, 2015). As a consequence, military-style networks have become high-value targets for cyber attacks, requiring strong cybersecurity controls, governance mechanisms, and continuous monitoring capabilities.

Over the past two decades, military and defence organisations have invested heavily in technical cybersecurity controls, including firewalls, intrusion detection systems, encryption mechanisms, and access control technologies (Johnson, 2015; U.S. Department of Defense, 2021). While these controls are essential, cyber incidents have continued to occur with regularity. A significant proportion of such incidents are not caused by technical system failures, but rather by human-layer cybersecurity policy violations. These include credential misuse, inappropriate privilege usage, unsafe endpoint behaviour such as unauthorised removable media insertion, and non-compliance with established security procedures (Bishop & Gates, 2008; Pfleeger & Caputo, 2012; Greitzer et al., 2014).

The human layer therefore, has emerged as a critical cybersecurity attack surface. Studies in behavioural and socio-technical cybersecurity have consistently demonstrated that user behaviour, organisational culture, leadership practices, and operational pressure strongly influence cybersecurity outcomes (Stanton et al., 2005; Sommestad et al., 2014; Parsons et al., 2017). In military environments, where discipline, hierarchy, and operational tempo shape day-to-day activity, human-layer policy violations can enable attack vectors such as credential compromise, privilege escalation, data exfiltration, and social engineering (Mitnick & Simon, 2011; Hadnagy, 2018; Krasznay & Hámornik, 2019).

Traditional approaches to managing human-layer cyber risk in military organisations rely primarily on policy dissemination, periodic awareness training, compliance attestations, and post-incident investigation. While it is necessary, these approaches provide limited capability for continuous detection, prioritisation, and measurable enforcement of cybersecurity policy compliance (Tsohou et al., 2015; ENISA, 2017). As a result, the human layer is often treated as a training problem rather than as an operationally measurable cybersecurity control surface.

Recent advances in artificial intelligence and machine learning have demonstrated effectiveness in intrusion detection, anomaly detection, and threat intelligence analysis (Buczak & Guven, 2016; Khraisat et al., 2019). However, the application of these techniques to detect human-layer cybersecurity policy violations remains limited. This limitation is particularly evident in military-style environments, where explainability, auditability, governance, and controlled automation are essential requirements (Sommer & Paxson, 2010; Karaman & Aybar, 2016).

These conditions create an opportunity for the development of constrained, technically grounded frameworks that integrate rule-based policy enforcement with modest, explainable machine learning techniques. Such hybrid approaches offer the potential to treat human-layer behaviour as a measurable and auditable cybersecurity signal, rather than as an abstract or purely behavioural concern (Greitzer et al., 2014; Sarker et al., 2020).

1.2 Research Problem Statement

Despite the existence of formal cybersecurity policies and established technical security controls, military organisations continue to experience cyber risk arising from human-layer cybersecurity policy violations. These violations include credential misuse, inappropriate privilege usage, unsafe endpoint interactions, and failure to comply with prescribed security procedures (Bishop & Gates, 2008; Pfleeger & Caputo, 2012).

Current approaches for managing these risks rely largely on awareness training, manual audits, and post-incident investigation. While these mechanisms have been support baseline compliance, they provide limited capability for continuous detection, prioritisation, and measurable enforcement of human-layer cybersecurity policies (Sommestad et al., 2014; Parsons et al., 2017). Consequently, cybersecurity teams often lack timely visibility into emerging human-layer risks and must rely on reactive responses after incidents have occurred.

Existing cybersecurity monitoring solutions predominantly focus on network- and system-level threats and offer limited mechanisms for treating human behaviour as a measurable and auditable cybersecurity attack surface (Sommer & Paxson, 2010; Buczak & Guven, 2016). Although artificial intelligence and machine learning techniques have been widely applied to technical anomaly detection, their use in detecting human-layer cybersecurity policy violations remains underdeveloped, particularly in military-style environments that demand explainability, governance, and strict control over automated decision-making (Greitzer et al., 2014; Karaman & Aybar, 2016).

As a result, there is a lack of empirically evaluated frameworks that combine rule-based policy enforcement with machine learning-based risk scoring to support the systematic detection of

human-layer cybersecurity policy violations. Without such frameworks, military organisations lack evidence-based mechanisms to assess the effectiveness of human-layer cybersecurity controls and to support informed cybersecurity governance decisions.

1.3 Research Aim and Objectives

1.3.1 Research Aim

The aim of this study is to develop and evaluate a hybrid rule-based and machine learning framework for detecting human-layer cybersecurity policy violations in military-style networks, using constrained telemetry, explainable analytics, and lab-based or synthetic evaluation methods.

1.3.2 Specific Research Objectives

The specific objectives of the study are to:

- i. Identify and operationally define a limited set of high-value human-layer cybersecurity policy violations relevant to military-style networks.
- ii. Design a rule-based policy model that encodes selected human-layer cybersecurity controls in a machine-checkable form.
- iii. Develop a machine learning-based risk scoring component that analyses behavioural telemetry to detect anomalous or potentially non-compliant user behaviour.
- iv. Integrate rule-based detection and machine learning risk scoring into a unified hybrid detection framework.
- v. Evaluate the framework using a controlled lab environment and/or synthetic datasets, based on detection accuracy, false positive rates, explainability, and operational feasibility.

1.4 Research Questions

The study is guided by the following research questions:

- i. Which human-layer cybersecurity policy violations can be operationally defined and detected within military-style networks using available telemetry?
- ii. How can rule-based policy enforcement be used to detect explicit human-layer cybersecurity policy violations?
- iii. How can machine learning techniques be applied to behavioural telemetry to identify anomalous or risky user behaviour related to policy non-compliance?
- iv. How effective is a hybrid rule-based and machine learning framework in detecting human-layer cybersecurity policy violations in a controlled evaluation environment?

- v. What are the strengths and limitations of the proposed framework in terms of accuracy, explainability, and governance requirements?

1.5 Definition of Key Terms

- **Human-Layer Cybersecurity:** The aspect of cybersecurity concerned with human behaviour, decision-making, and compliance as a measurable component of security systems (Pfleege & Caputo, 2012).
- **Cybersecurity Policy Violation:** Any action or behaviour that deviates from formally defined cybersecurity policies, procedures, or controls.
- **Hybrid Detection Framework:** A detection approach that integrates deterministic rule-based logic with machine learning-based analytics.
- **Hybrid Rule-Based:** It is the integration of predefined cybersecurity policy and behavioural rules with machine learning modules to detect, validate and adaptively identify human-layer policy violations within a controlled military-style network environment.
- **Machine Learning Risk Scoring:** The use of statistical or learning-based models to assign relative risk levels to observed behaviours based on deviation from expected patterns (Greitzer et al., 2014).
- **Military-Style Networks:** Network environments characterised by strict access control, auditability, hierarchical governance, and operational security constraints (Johnson, 2015; Karaman & Aybar, 2016).

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

Cybersecurity research has traditionally focused on protecting systems, networks, and data from external technical threats. However, as cyber defence capabilities have matured, it has become increasingly clear that human behaviour within secure environments remains a persistent and significant source of cyber risk. This chapter reviews existing research relevant to human-layer cybersecurity, policy compliance, and the application of rule-based and machine learning techniques for detection within security-critical environments.

The review is structured to support the study's aim of developing and evaluating a hybrid rule-based and machine learning framework for detecting human-layer cybersecurity policy violations in military-style networks. It first examines prior research on human-layer cybersecurity risks and policy compliance. It then reviews technical approaches to detecting such risks, including rule-based enforcement and machine learning-based anomaly detection. The chapter concludes by identifying gaps in the literature and presenting the theoretical and conceptual framework underpinning the study.

2.2 Review of Existing Research

2.2.1 Human-Layer Cybersecurity and Policy Compliance

The human layer has long been recognised as a critical factor in cybersecurity effectiveness. Early work by Pfleeger and Caputo (2012) and Stanton et al. (2005) demonstrated that user behaviour, decision-making, and compliance with security policies significantly influence organisational cyber resilience. Rather than being purely accidental, many security incidents arise from predictable patterns of non-compliance, workload pressure, and misaligned incentives.

In military and defence contexts, human-layer cybersecurity risks are amplified by hierarchical command structures, operational urgency, and strict access control requirements. Bishop and Gates (2008) identify insider-related cyber risk as a major threat vector in high-security environments, noting that both malicious and unintentional actions can result in severe consequences. Greitzer et al. (2014) further argue that insider-related cyber risk must be treated as a continuous monitoring problem rather than a post-incident investigative task.

Cybersecurity policy compliance has traditionally been addressed through awareness training, policy documentation, and disciplinary measures. While these mechanisms are necessary, research indicates that training alone does not reliably translate into compliant behaviour, particularly under operational pressure (Sommestad et al., 2014; Parsons et al., 2017). Tsohou et al. (2015) and

ENISA (2017) observe that compliance programmes often lack measurable feedback mechanisms, making it difficult to assess their effectiveness or identify emerging risk patterns.

These limitations highlight the need to treat human-layer compliance not merely as a behavioural issue, but as an operational cybersecurity control that can be monitored, measured, and evaluated using technical means.

2.2.2 Rule-Based Approaches to Cybersecurity Policy Enforcement

Rule-based systems have a long history in cybersecurity, particularly in access control, intrusion detection, and policy enforcement. Deterministic rule engines offer transparency, predictability, and auditability, making them well suited to environments with strict governance requirements (Rot, 2008; Sommer & Paxson, 2010).

In defence and military-style networks, rule-based enforcement aligns with established doctrines of accountability, least privilege, and procedural compliance (Johnson, 2015; Karaman & Aybar, 2016). Security policies can be formalised as machine-checkable rules that define acceptable behaviour, prohibited actions, and required evidence of compliance. This approach is often referred to as “policy-as-code” and enables consistent enforcement and auditing of security controls (U.S. Department of Defense, 2021).

However, rule-based systems alone are limited in their ability to detect subtle or emerging behavioural risks. They are effective at identifying explicit policy violations, such as unauthorised privilege escalation or prohibited device usage, but struggle with ambiguous or context-dependent behaviours (Greitzer et al., 2014). As a result, purely rule-based approaches may generate high false-positive rates or fail to identify early indicators of insider-related risk.

2.2.3 Machine Learning for Behavioural and Anomaly Detection

Machine learning has been widely applied in cybersecurity for intrusion detection, malware classification, and anomaly detection. Surveys by Buczak and Guven (2016) and Khraisat et al. (2019) highlight the effectiveness of machine learning in identifying deviations from normal system behaviour using large volumes of telemetry data.

In the context of human-layer cybersecurity, machine learning has primarily been explored for insider threat detection and user behaviour analytics. Greitzer et al. (2014) demonstrate that behavioural indicators such as login timing, access frequency, and privilege usage can provide early warning signals of elevated risk. Sarker et al. (2020) argue that machine learning is particularly well suited for modelling complex behavioural patterns that are difficult to capture through deterministic rules.

Despite these advantages, the application of machine learning in security-critical environments faces significant challenges. Sommer and Paxson (2010) caution against over-reliance on complex models that lack explainability, noting that opaque detection mechanisms undermine trust and operational adoption. In military contexts, where automated decisions must be auditable and defensible, explainability and governance are often more important than marginal gains in detection accuracy (Karaman & Aybar, 2016).

These constraints suggest that machine learning should be applied in a limited and supportive role, focused on risk scoring and anomaly prioritisation rather than autonomous decision-making.

2.2.4 Hybrid Rule-Based and Machine Learning Approaches

Recent research increasingly supports hybrid cybersecurity architectures that combine rule-based enforcement with machine learning-based analytics. Such approaches aim to leverage the strengths of both techniques while mitigating their individual limitations (Greitzer et al., 2014)

In hybrid frameworks, rule-based components are typically used to enforce explicit policy requirements and detect clear violations, while machine learning models analyse behavioural telemetry to identify anomalous or weak signals that may warrant further investigation. This division of labour supports explainability, reduces false positives, and aligns with governance requirements in high-assurance environments (Leonhardi & Murphy, 2015; Sarker et al., 2020).

However, existing hybrid approaches are often evaluated in civilian or enterprise contexts and rarely focus explicitly on human-layer cybersecurity policy violations within military-style networks. Furthermore, many studies lack rigorous evaluation using measurable, system-generated evidence, relying instead on conceptual models or limited datasets.

2.2.5 Cybersecurity in Military and High-Assurance Environments

Military networks differ from civilian environments due to classification requirements, strict access controls, operational secrecy, and governance obligations. Healey (2013) and Johnson (2015) highlight that military cybersecurity must prioritise resilience, auditability, and mission assurance over convenience or automation.

African and developing-country military contexts face additional challenges, including limited access to advanced security tooling and skilled personnel (Simfukwe & Shemi, 2024; Bester & Arendse, 2024). These constraints necessitate cybersecurity solutions that are efficient, explainable, and feasible within constrained environments, further motivating the need for a scoped and practical framework.

2.3 Identification of Research Gaps

The review of existing literature reveals several critical gaps:

- i. **Human-layer cybersecurity is under-operationalised:** While human behaviour is widely acknowledged as a major source of cyber risk, most existing approaches rely on training and policy dissemination rather than technical detection and enforcement mechanisms (Parsons et al., 2017; Tsohou et al., 2015).
- ii. **Limited integration of policy enforcement and behavioural detection:** Existing compliance systems focus on documentation and audits, while detection systems focus on technical threats. Few frameworks integrate policy-as-code enforcement with behavioural risk detection (Leonhardi & Murphy, 2015).
- iii. **Underexplored application of ML to human-layer policy violations:** Although ML has been widely applied to intrusion detection, its use in detecting human-layer cybersecurity policy violations remains limited, particularly in constrained military-style environments (Buczak & Guven, 2016; Sarker et al., 2020).
- iv. **Lack of explainable, governance-aligned frameworks for military contexts:** Military environments require explainability, auditability, and strict control over automated decision-making. Existing ML-heavy approaches often fail to meet these requirements (Sommer & Paxson, 2010; Johnson, 2015).

These gaps justify the need for a hybrid rule-based and machine learning framework that supports the measurable detection of human-layer cybersecurity policy violations in military-style

2.4 Theoretical and Conceptual Framework

This study is grounded in socio-technical systems theory, which views cybersecurity as the outcome of interactions between technical systems, human behaviour, and organisational controls (Stanton et al., 2005; Sommestad et al., 2014). Within this perspective, human actions are not external to the system but form an integral part of the security control surface.

The research also draws on design science research principles, which emphasise the development and evaluation of artefacts that address real-world security problems (Rot, 2008). This aligns with the study's focus on building and evaluating a practical detection framework rather than conducting purely conceptual analysis.

Conceptually, the proposed framework integrates four core components:

- i. **Policy and Control Layer:** Human-layer cybersecurity policies are formalised into machine-checkable rules representing acceptable and prohibited behaviours.
- ii. **Telemetry and Evidence Layer:** Human-related cyber events (e.g., authentication activity, privilege usage, endpoint interactions) are collected from logs or synthetic datasets.
- iii. **Detection and Risk Scoring Layer:** Rule-based mechanisms detect explicit policy violations, while machine learning models perform anomaly-based risk scoring on behavioural patterns.
- iv. **Governance and Response Layer:** Detected violations and risk indicators support alerts, reports, and governance decisions, maintaining explainability and auditability.

This framework reflects a hybrid detection paradigm in which deterministic rules provide governance and auditability, while machine learning supports adaptability and prioritisation. By grounding the framework in measurable, system-generated evidence, the study aligns with the operational realities and constraints of military-style networks.

CHAPTER THREE: RESEARCH DESIGN AND METHODOLOGY

3.1 Research Approach

This study adopts a design science research (DSR) approach to develop and empirically evaluate a hybrid rule-based and machine learning framework for detecting human-layer cybersecurity policy violations in military-style networks. Design science research is appropriate where the primary contribution is a technical artefact designed to address a clearly defined real-world problem, supported by systematic evaluation using measurable criteria (Rot, 2008).

The research focuses on the design, implementation, and evaluation of a constrained prototype framework, rather than on organisational change or behavioural intervention. The human layer is treated explicitly as a cybersecurity signal source, not as a subject of psychological or sociological analysis. Human behaviour is operationalised through system-generated telemetry, such as authentication events, privilege usage, and endpoint interactions.

A mixed-methods strategy is employed in a supporting role. Qualitative inputs are used strictly for system requirement validation and rule definition, while quantitative methods are used for model training, detection performance measurement, and framework evaluation. This approach is consistent with applied cybersecurity research practices in high-assurance environments (Akraminia, 2024).

3.2 Framework Overview and Scope Definition

To control scope and ensure feasibility, the framework targets a limited set of high-value human-layer cybersecurity policy violations, selected based on prevalence in the literature and suitability for system-based detection. This control set informs both the rule-based and machine learning components of the framework, providing measurable, auditable signals for policy violation detection. These include:

- Credential misuse indicators (e.g., repeated failed logins, anomalous login timing),
- Inappropriate privilege usage (e.g., abnormal administrative access frequency),
- Unsafe endpoint interactions (e.g., unauthorised removable media usage),
- Non-compliance with mandatory cybersecurity controls (e.g., incomplete policy training or attestations).

The framework consists of three tightly coupled layers, each of which is directly reflected in the research methodology:

- i. **Rule-Based Policy Layer:** deterministic detection of explicit policy violations. The operational definition of the human-layer policy controls, including credential use, privileged access, endpoint interactions, and mandatory policy compliance, is provided in **Appendix C**.
- ii. **Machine Learning Risk Scoring Layer:** identification of anomalous or elevated-risk behaviour,
- iii. **Hybrid Detection and Evaluation Layer:** integration and empirical assessment of detection outcomes.

3.3 Data Collection Methods

Data collection is conducted exclusively for technical system development and evaluation purposes. No classified, operational, or personally identifiable military data are collected.

3.3.1 Policy and Control Documentation Analysis

Cybersecurity policies, acceptable-use policies, and control guidelines relevant to military-style networks are analysed to extract:

- Control objectives,
- Prohibited actions,
- Required evidence of compliance,
- Thresholds for violation detection.

These artefacts form the basis of the rule-based policy layer, ensuring that detection logic is grounded in formally defined security requirements (U.S. Department of Defense, 2021; British Army, 2020; SANDF, 2020).

3.3.2 Expert-Guided System Requirement Validation

Semi-structured technical interviews are conducted with cybersecurity practitioners, including security analysts and policy administrators. These interviews are used to:

- Validate selected human-layer violation categories,
- Refine rule logic and thresholds,
- Confirm operational realism of simulated scenarios.

Expert consultation is widely accepted in cybersecurity design research for requirement validation and threat modelling (Leonhardi & Murphy, 2015).

3.3.3 Synthetic and Lab-Generated Telemetry Data

Given access constraints in military environments, the study employs:

- Synthetic datasets,
- Lab-generated authentication and endpoint logs,
- Scenario-based user behaviour simulations.

The proposed evaluation of the hybrid framework relies exclusively on synthetic and lab-generated datasets, detailed in **Appendix B** and **Appendix D**. These datasets are designed to reflect realistic distributions of normal and policy-violating behaviour while avoiding operational risk and ethical concerns. Synthetic data usage is consistent with best practices in AI-enabled cybersecurity research (Akraminia, 2024).

3.4 Sampling Strategy

3.4.1 Expert Input Sampling

A purposive sampling strategy is used to select experts with direct experience in cybersecurity operations, compliance, or policy enforcement. Participants are chosen based on technical relevance rather than representativeness, ensuring system-focused input (Rot, 2008).

3.4.2 System Evaluation Data Sampling

No human participants are sampled for behavioural observation. All behavioural inputs are derived from synthetic or anonymised system logs generated within a controlled lab environment.

3.5 Data Analysis and Framework Implementation

3.5.1 Rule-Based Policy Detection

Extracted policies are formalised into:

- If-then detection rules,
- Threshold-based compliance checks,
- Deterministic violation indicators.

This layer detects explicit policy violations and provides explainable, auditable detection outputs aligned with military governance requirements (SANDF, 2020; U.S. Department of Defense, 2021).

3.5.2 Machine Learning Risk Scoring

Machine learning techniques are applied to behavioural telemetry to identify anomalous or elevated-risk user behaviour. The models focus on:

- Anomaly detection rather than autonomous decision-making,
- Feature sets derived from authentication and endpoint events,
- Risk scoring outputs to support prioritisation.

Model performance is evaluated using standard cybersecurity metrics, including precision, recall, false-positive rate, and detection latency (Buczak & Guven, 2016; Akraminia, 2024).

3.5.3 Hybrid Detection Integration

Outputs from the rule-based and machine learning layers are combined to produce a unified detection outcome. Rule violations represent confirmed policy breaches, while machine learning scores indicate potential risk requiring further investigation. This hybrid approach balances governance, explainability, and adaptability (Greitzer et al., 2014; Sarker et al., 2020).

3.6 Evaluation Strategy and Plan

The framework is evaluated using scenario-based testing within a controlled environment to reflect the realistic policy violation cases reported in the literature. Evaluation criteria include:

- Detection accuracy and false-positive rates,
- Ability to detect policy violations under varying behavioural scenarios,
- Explainability and auditability of detection outputs,
- Operational feasibility in military-style network constraints.

This evaluation strategy ensures measurable, system-generated evidence without reliance on classified infrastructure.

To ensure full traceability between research objectives, questions, and system components, a mapping of human-layer policy controls to research questions and framework layers is presented in **Appendix E**. This table illustrates how each control contributes to measurable detection outcomes and supports empirical evaluation of the hybrid framework.

3.7 Ethical Considerations

3.7.1 Data Protection and Privacy

- No real military operational data are collected.
- All datasets are synthetic or anonymised.
- No personal identifiers are processed or stored.

3.7.2 Informed Consent

Expert participants are to be informed of the technical purpose of the study and provide written consent prior to participation.

3.7.3 Avoidance of Harm

The study explicitly avoids surveillance of individuals, intelligence gathering, or monitoring of real military operations. Ethical safeguards align with defence research guidelines (SANDF, 2020; U.S. Department of Defense, 2021).

3.7.4 Ethical Approval

Ethical clearance is obtained from the university, and institutional permission maybe sought where expert consultation involves defence-related personnel.

CHAPTER FOUR: SIGNIFICANCE OF THE STUDY

4.1 Contribution to Knowledge

This study makes several contributions to knowledge in the field of cybersecurity, particularly in relation to human-layer risk detection in military-style networks.

4.1.1 Operationalisation of Human-Layer Cybersecurity Policy Violations

The study advances human-centric cybersecurity theory by operationalising the human layer as a measurable and auditable component of cybersecurity systems, rather than treating it solely as a behavioural or awareness-related issue. While prior research has established that human behaviour significantly influences cybersecurity outcomes, much of this work remains conceptual or training-focused (Pfleeger & Caputo, 2012; Sommestad et al., 2014). This research contributes by demonstrating how selected human-layer cybersecurity policy violations can be translated into machine-detectable events derived from system-generated evidence.

4.1.2 Contribution to Hybrid Detection Approaches in Cybersecurity

The study contributes to AI-enabled cybersecurity scholarship by demonstrating the application of a hybrid rule-based and machine learning detection framework to human-layer cybersecurity policy violations. While machine learning has been widely applied to intrusion detection and malware analysis, its use in human-layer compliance detection remains limited (Buczak & Guven, 2016; Sarker et al., 2020). By combining deterministic rule-based enforcement with machine learning-based risk scoring, the study provides a balanced detection approach that supports explainability, auditability, and governance which are key requirements in security-critical environments (Sommer & Paxson, 2010).

4.1.3 Methodological Contribution to Military Cybersecurity Research

The study contributes methodologically by demonstrating how lab-based and synthetic datasets can be used to evaluate human-layer cybersecurity detection mechanisms in the absence of access to classified or operational military networks. This approach provides a practical pathway for empirical validation in constrained military environments and addresses a gap in existing research that often lacks measurable evaluation strategies (Greitzer et al., 2014; Akraminia, 2024).

4.1.4 Contextual Contribution to Military-Style Network Security

By focusing explicitly on military-style networks, the study enriches cybersecurity literature that is predominantly centred on civilian or enterprise contexts. The framework reflects operational characteristics such as strict access control, hierarchical governance, and auditability requirements, contributing context-specific insights relevant to defence and security organisations (Johnson, 2015; Karaman & Aybar, 2016).

4.2 Practical Implications

In addition to its theoretical contributions, the study offers several practical implications for cybersecurity operations and governance in military-style environments.

4.2.1 Support for Continuous Detection of Human-Layer Policy Violations

The proposed framework provides a practical mechanism for the continuous detection of selected human-layer cybersecurity policy violations, moving beyond reliance on periodic training, manual audits, and post-incident investigations. By leveraging system-generated telemetry, the framework enables earlier identification of risky or non-compliant behaviour and supports proactive cyber risk management (Greitzer et al., 2014; Parsons et al., 2017).

4.2.2 Enhancement of Evidence-Based Cybersecurity Governance

The framework supports evidence-based cybersecurity governance by providing measurable indicators of policy compliance derived from technical systems. This enables cybersecurity managers and commanders to make informed decisions based on observable risk patterns rather than assumptions or retrospective analysis, aligning with military doctrines emphasising discipline, accountability, and operational readiness (Johnson, 2015; Krasznay & Hámornik, 2019).

4.2.3 Improved Policy Enforcement and Compliance Assurance

By integrating rule-based policy detection with machine learning-based risk scoring, the framework enhances the enforcement of existing cybersecurity policies and standards. It complements established governance frameworks such as ISO/IEC 27001 and the NIST Cybersecurity Framework by operationalising the human layer as an enforceable control domain rather than a passive compliance requirement (NIST, 2020; ISO/IEC 27001, 2022).

4.2.4 Applicability to Other High-Security Domains

Although the study is situated within military-style networks, the constrained and explainable nature of the framework makes it adaptable to other high-security environments, including law enforcement agencies, intelligence organisations, and critical national infrastructure operators. The framework therefore provides a foundation for broader application and future research across security-sensitive sectors.

4.3 Concluding Statement

Overall, the significance of this study lies in its ability to demonstrate that human-layer cybersecurity policy violations can be detected, measured, and evaluated using a constrained, technically grounded framework.

CHAPTER FIVE: RESEARCH PLAN AND TIMELINE

5.1 Project Phases

The research will be conducted over a period of ten (10) months, structured into six sequential and partially overlapping phases, as illustrated in Appendix A (Research Timeline / Gantt Chart). Each phase is designed to progressively build towards the design, implementation, and evaluation of the proposed hybrid rule-based and machine learning framework.

5.1.1 Phase 1: Literature Review and Problem Refinement (Months 1–2)

The first phase focuses on conducting a comprehensive review of existing literature related to human-layer cybersecurity, policy compliance monitoring, insider threat detection, rule-based security controls, and machine learning techniques for behavioural analysis. This phase aims to refine the research problem, identify gaps in existing approaches, and establish the theoretical and conceptual foundations of the study.

Key activities include:

- Reviewing academic journals, conference papers, and industry standards relevant to human-layer cybersecurity and compliance monitoring.
- Analysing existing rule-based and machine learning approaches used in security monitoring systems.
- Refining the research objectives, questions, and system scope based on identified gaps.

Outputs:

Refined research problem, finalised objectives and research questions, and a structured literature review.

5.1.2 Phase 2: System Requirements Analysis and Framework Design (Month 3)

This phase involves translating the research objectives into technical system requirements and designing the proposed hybrid framework. The focus is on defining the scope of human-layer policy violations, identifying relevant security controls, and designing the overall system architecture.

Key activities include:

- Defining the selected human-layer behaviours and policy controls to be monitored.
- Designing the hybrid framework architecture, including rule-based detection, machine learning-based risk scoring, and enforcement workflow components.

- Developing data flow models and defining system inputs and outputs.

Outputs:

System requirements specification, conceptual and architectural framework diagrams (Appendix F and G).

5.1.3 Phase 3: Laboratory Testbed Development and Data Preparation (Months 4–5)

In this phase, a controlled laboratory testbed representing a military-style network environment will be developed. The testbed will support the generation and collection of realistic synthetic data required for system evaluation.

Key activities include:

- Setting up a virtualised network environment with authentication services, user workstations, and centralised logging.
- Configuring telemetry sources such as authentication logs, privilege usage logs, and endpoint activity logs.
- Generating realistic synthetic datasets through scripted user behaviours and policy violation scenarios.
- Labelling collected data to establish ground truth for evaluation purposes.

Outputs:

Operational laboratory testbed, centralised logging infrastructure, and labelled synthetic datasets.

5.1.4 Phase 4: Framework Implementation (Months 6–7)

This phase focuses on implementing the core components of the proposed hybrid framework. Both the rule-based compliance detection layer and the machine learning-based risk scoring module will be developed and integrated.

Key activities include:

- Implementing rule-based policy violation detection mechanisms.
- Developing and training machine learning models for anomaly-based user risk scoring.
- Integrating rule-based outputs with machine learning results to support hybrid decision-making.
- Implementing alerting, reporting, and workflow mechanisms.

Outputs:

Fully implemented hybrid detection framework and integrated system prototype.

5.1.5 Phase 5: Experimental Evaluation and Data Analysis (Month 8)

The fifth phase involves evaluating the performance and effectiveness of the proposed framework using controlled experimental scenarios within the laboratory testbed.

Key activities include:

- Executing predefined user behaviour and policy violation scenarios.
- Measuring system performance using metrics such as detection accuracy, false positive rate, detection latency, and risk scoring effectiveness.
- Analysing and interpreting experimental results in relation to the research objectives.

Outputs:

Evaluation results, performance metrics, and analysed experimental findings.

5.1.6 Phase 6: Dissertation Writing and Final Revisions (Months 9–10)

The final phase focuses on documenting the research process and results in the form of a complete dissertation.

Key activities include:

- Writing and refining dissertation chapters.
- Integrating evaluation results and discussion of findings.
- Addressing limitations and outlining recommendations for future work.
- Final proofreading and formatting in accordance with institutional guidelines.

Outputs:

Final dissertation submitted for examination.

A detailed timeline showing the duration, sequencing, and milestones of each project phase is provided in **Appendix A**.

5.2 Research Timeline

The research timeline spans twelve (10) months and is summarised in **Table 5.1** below, which presents the distribution of activities across the project duration. A detailed research timeline outlining all project phases, activities, deliverables, and milestones is provided in **Appendix A**.

Table 5.1: Summary of Research Timeline (10 Months)

Research Phase	Months
Problem Refinement & Literature Review	1–2
System Requirements Analysis and Framework Design	3
Laboratory Testbed Development and Data Preparation	4–5
Framework Implementation	6–7
Experimental Evaluation and Data Analysis	8
Dissertation Writing, Final Revisions and Submission	9–10

5.3 Concluding Statement

This research plan and timeline demonstrate a feasible, well-scoped approach to developing and evaluating a hybrid rule-based and machine learning framework for detecting human-layer cybersecurity policy violations. The phased structure ensures that each activity contributes directly to the research aim, supports measurable evaluation, and adheres to the constraints and expectations of military-style cybersecurity research.

CHAPTER SIX: EXPECTED OUTCOMES

6.1 Anticipated Results

This chapter outlines the expected outcomes of the study, focusing on the anticipated technical results and the expected contribution to knowledge in the field of cybersecurity. The outcomes are framed to reflect realistic achievements within the research context and are aligned with the objectives of designing and evaluating a hybrid rule-based and machine learning framework for detecting human-layer policy violations in military-style networks.

6.1.1 Demonstration of a Feasible and Workable Prototype System

The study is expected to demonstrate the feasibility of implementing an end-to-end, prototype detection framework within a constrained, non-classified lab environment. The outcome will show that a technically grounded system comprising policy rules, telemetry collection, hybrid detection logic, and evaluation mechanisms can be built and evaluated within a timeframe.

6.1.2 Detection of Human-Layer Cybersecurity Policy Violations

This study is expected to demonstrate that human-layer cybersecurity policy violations in military-style networks can be systematically detected using a hybrid rule-based and machine learning framework. The framework is anticipated to identify high-risk behaviours such as credential misuse, inappropriate privilege usage, unsafe endpoint interactions, and non-compliance with mandatory security controls through continuous analysis of system-generated telemetry.

6.1.3 Improved Accuracy through Hybrid Rule-Based and Machine Learning Detection

The hybrid detection approach is expected to outperform rule-only mechanisms by combining deterministic policy enforcement with machine learning-based risk scoring. Rule-based logic is anticipated to reliably detect explicit policy violations, while the machine learning component is expected to identify anomalous or weak-signal behaviours that may not breach explicit rules but indicate elevated risk. This hybrid approach is expected to reduce false positives while improving detection coverage.

6.1.4 Explainable and Auditable Detection Outcomes

Given the governance and accountability requirements of military-style environments, the proposed framework is expected to produce explainable and auditable detection outcomes. Rule violations are expected to be traceable to specific policy definitions, while machine learning outputs are anticipated to include interpretable risk indicators and contributing features. This outcome supports trust, oversight, and human-in-the-loop decision-making.

6.1.5 Measurable Cybersecurity Compliance Evidence

The framework is expected to generate measurable, system-derived evidence of human-layer cybersecurity compliance. This includes metrics such as detected policy violations, risk scores, response actions, and compliance trends over time. Such evidence is expected to support informed cybersecurity governance, auditing, and continuous improvement without reliance on self-reported behaviour or post-incident analysis alone.

6.1.6 Contribution to Military-Style Cybersecurity Governance and Decision-Making

The proposed framework is expected to contribute practical insights into how human-layer cybersecurity controls can be operationalised within military-style networks. By providing timely detection, risk prioritisation, and evidence-based reporting, the framework is anticipated to support more informed decision-making related to access control, monitoring priorities, and policy enforcement.

6.1.7 Foundation for Future Research and System Extension

Finally, the study is expected to provide a reusable conceptual and technical foundation for future research. The framework may be extended to additional human-layer behaviours, integrated with other security controls, or adapted for use in other high-assurance environments such as law enforcement, intelligence services, or critical national infrastructure.

6.2 Concluding Statement

Overall, the expected outcomes of this study are intentionally modest and measurable. Rather than claiming to eliminate insider threats or enforce comprehensive compliance, the study is expected to demonstrate that a constrained hybrid detection framework can provide meaningful, explainable, and empirically validated insights into human-layer cybersecurity policy violations in military-style networks.

CHAPTER SEVEN: ESTIMATED BUDGET AND RESOURCES

7.1 Estimated Budget

This study is designed to be conducted within a controlled laboratory and simulation environment, without reliance on classified infrastructure, proprietary military systems, or restricted datasets. As such, the estimated budget focuses primarily on computing resources, software tools, data generation environments, and documentation support, ensuring that the research remains feasible within a timeframe and funding context.

The proposed hybrid framework will be implemented using open-source technologies and standard computing equipment, minimising financial cost while maintaining technical rigour. Where possible, institutional resources available through the university will be utilised.

7.2 Hardware Resources

The hardware requirements for this study are modest and are expected to be met through either existing personal equipment or university laboratory facilities.

The primary hardware resources include:

- A standard personal computer or laptop with sufficient processing capability to support virtualisation and model training (minimum 16 GB RAM, multi-core CPU).
- Optional use of an external monitor to support development, monitoring, and evaluation activities.
- No specialised hardware (e.g., GPUs or dedicated servers) is required due to the use of lightweight and explainable machine learning models.

These hardware resources are expected to support the deployment of virtual machines, log generation, data processing, and performance evaluation tasks required by the study.

7.3 Software and Development Tools

The study will rely primarily on open-source and freely available software, reducing licensing costs and ensuring reproducibility.

Key software resources include:

- Virtualisation platforms (e.g., VirtualBox or VMware Workstation) to simulate military-style network environments.
- Operating systems such as Linux distributions and Windows Server for identity and endpoint simulation.

- Programming languages and development environments, including Python and associated libraries for data processing and machine learning.
- Log aggregation and analysis tools (e.g., ELK Stack or equivalent open-source platforms).
- Version control systems (e.g., Git) for source code management.
- Documentation tools (e.g., Microsoft Word or LaTeX) for thesis preparation.

All selected software tools are expected to be available at no cost or through existing academic licenses.

7.4 Data Resources

Given the sensitivity of military cybersecurity data, the study will not depend on real operational datasets. Instead, data resources will be generated through controlled laboratory simulations and synthetic data generation.

Data resources will include:

- Authentication and access control logs generated from a simulated directory service environment.
- Endpoint activity logs representing user interactions with systems and removable media.
- Policy compliance and violation records generated through scripted scenarios.
- Synthetic datasets designed to model both compliant and non-compliant user behaviour patterns.

This approach ensures ethical compliance, avoids classified data exposure, and supports repeatable evaluation.

7.5 Human and Institutional Resources

The primary human resource for this study is the researcher, who will be responsible for system design, implementation, experimentation, analysis, and documentation.

Additional institutional resources include:

- Academic supervision for methodological guidance and quality assurance.
- Access to university computing laboratories and library services.

7.6 Estimated Budget Summary

Table 7.1 below presents a summary of the estimated budget for the study.

Item	Description	Estimated Cost (ZMW)
Hardware	Personal computer	Existing
Software	Open-source tools and academic licenses	No additional cost
Virtualisation	Virtual machines and test environments	No additional cost
Data Generation	Synthetic and lab-generated datasets	No additional cost
Documentation	Printing and binding of thesis	3,000
Contingency	Minor unforeseen expenses	2,000
Total		5,000

7.7 Cost Management and Sustainability

Cost management will be achieved by prioritising open-source solutions, reusing existing infrastructure, and avoiding unnecessary technological complexity. The selected tools and environments are widely supported and sustainable, enabling future extension or replication of the study without additional financial burden.

CHAPTER EIGHT: CONCLUSION

This research proposal has outlined a focused and technically grounded study aimed at addressing human-layer cybersecurity policy violations within military-style networks. Recognising that a significant proportion of cyber risk in security-critical environments arises from human behaviour rather than purely technical system failures, the study positions the human layer as a measurable and auditable component of cybersecurity governance.

The proposed research responds directly to limitations identified in existing approaches to human-layer cybersecurity management, which rely heavily on awareness training, policy dissemination, and post-incident investigation. While these measures remain important, they provide limited capability for continuous detection and evidence-based enforcement of cybersecurity policies. By contrast, the proposed study advances a constrained hybrid framework that integrates deterministic rule-based policy enforcement with explainable machine learning-based risk scoring, enabling the detection of selected human-layer cybersecurity policy violations using system-generated evidence.

The proposal has clearly defined the scope of the study, limiting it to a manageable set of high-value human-layer behaviours and avoiding overextension into broad insider threat or behavioural research. The methodology adopts a design science research approach, ensuring that the primary contribution is a buildable and testable technical artefact. Evaluation is planned using lab-based and synthetic datasets, addressing the practical and ethical constraints associated with military environments while maintaining empirical rigour.

By aligning technical design, detection logic, and evaluation metrics with the governance, auditability, and explainability requirements of military-style networks, the study demonstrates both academic defensibility and practical relevance. The anticipated outcomes of the research include empirical evidence on the effectiveness of hybrid detection approaches for human-layer cybersecurity policy violations, as well as a reusable framework that can inform future research and practice in other high-security contexts.

In summary, this proposal presents a feasible and well-scoped research plan that contributes to cybersecurity scholarship by operationalising the human layer as a detectable and measurable attack surface. Through its emphasis on constrained design, measurable evaluation, and alignment with real-world security requirements, the study is positioned to deliver meaningful insights and practical value within the limits of a master's level research project.

REFERENCES

- Akraminia, M. (2024). The cultural benefits and disadvantages of virtual social networks among military personnel. *Military Science and Tactics*, 20(67), 5-31.
- Anderson, R. (2008). *Security engineering: A guide to building dependable distributed systems*.
- Behl, A., & Behl, K. (2017). Cyberwar: The next threat to national security. *Journal of Financial Crime*, 24(3), 408–417.
- Bester, K., & Arendse, D. E. (2024). Measuring cybersecurity awareness in a South African military sample. *Scientia Militaria: South African Journal of Military Studies*, 52(1), 5-33.
- Bishop, M., & Gates, C. (2008). Defining the insider threat. *Proceedings of the 4th Annual Workshop on Cyber Security and Information Intelligence Research*.
- British Army. (2020). *Cyber and Electromagnetic Activities Doctrine*.
- Buczak, A. L., & Guven, E. (2016). A survey of data mining and machine learning methods for cybersecurity intrusion detection. *IEEE Communications Surveys & Tutorials*, 18(2), 1153–1176.
- ENISA. (2017). *Cybersecurity culture guidelines*.
- Greitzer, F. L., et al. (2014). Predictive modeling for insider threat mitigation. *IEEE Security & Privacy*, 12(6), 30–37.
- Hadnagy, C. (2018). *Social engineering: The science of human hacking*. Wiley.
- Healey, J. (2013). *A fierce domain: Conflict in cyberspace, 1986–2012*. Cyber Conflict Studies Association.
- ISO/IEC 27001. (2022). *Information security management systems*.
- Jain, A., Tailang, H., Goswami, H., Dutta, S., Sankhla, M. S., & Kumar, R. (2016). Social engineering: Hacking a human being through technology. *IOSR Journal of Computer Engineering*, 18(5), 94-100.
- Johnson, T. A. (Ed.). (2015). *Cybersecurity: Protecting critical infrastructures from cyber attack and cyber warfare*. CRC Press.
- Karaman, Muhammed, and Celil Aybar. "Institutional cybersecurity from military perspective." *International journal of information security science* 5, no. 1 (2016): 1-7.
- Khraisat, A., et al. (2019). Survey of intrusion detection systems. *IEEE Communications Surveys & Tutorials*, 21(2), 1412–1447.
- Krasznay, C., & Hámornik, B. P. (2019). Human Factors Approach to Cybersecurity Teamwork—The Military Perspective. *Advances in Military Technology*, 14(2), 291-305.
- Leonhardi Eric V, Mark Murphy “Analysis of Department of Defense Social Media Policy And its Impact On Operation Security,” 2015

- Mitnick, K. D., & Simon, W. L. (2011). *The art of deception*. Wiley
- NIST. (2020). *Cybersecurity Framework (CSF)*.
- Onukwuli, S. K., Okpala, C. C., Edeh, M. O., & Udu, C. (2025). Human-Centric Cybersecurity: The Integration of Psychological Insights and Socio-Technical Systems. *INTERNATIONAL JOURNAL OF INDUSTRIAL AND PRODUCTION ENGINEERING*, 3(4), 52-67.
- Parsons, K., et al. (2017). The human aspects of information security questionnaire (HAIS-Q). *Computers & Security*, 42, 165–176.
- Pfleeger, S. L., & Caputo, D. D. (2012). Leveraging behavioral science to mitigate cyber security risk. *Computers & Security*, 31(4), 597–611.
- Phiri, V. M., & Mupeta, M. (2025). AI-driven cybersecurity awareness systems. *International Journal of Information Security Science*.
- Rot, A. (2008, October). IT risk assessment: Quantitative and qualitative approach. In *proceedings of the World Congress on Engineering and Computer Science* (pp. 1073-1078).
- Sabev, S. I. (2020). Integrated approach to cyber defence: Human in the loop. technical evaluation report. *Information & Security: An International Journal*, 44, 76-92.
- Sarker, I. H., Kayes, A. S. M., Badsha, S., Alqahtani, H., Watters, P., & Ng, A. (2020). Cybersecurity data science: an overview from machine learning perspective. *Journal of Big data*, 7(1), 41.
- Shahana, A., Hasan, R., Farabi, S. F., Akter, J., Mahmud, M. A. A., Johora, F. T., & Suzer, G. (2024). AI-driven cybersecurity: Balancing advancements and safeguards. *Journal of Computer Science and Technology Studies*, 6(2), 76-85.
- Simfukwe, M., & Shemi, A. (2024). Cybersecurity maturity in African defence institutions. *African Security Review*.
- Sommer, R., & Paxson, V. (2010). Outside the closed world: On using machine learning for network intrusion detection. *IEEE Symposium on Security and Privacy*.
- Sommestad, T., Hallberg, J., Lundholm, K., & Bengtsson, J. (2014). Variables influencing information security policy compliance. *Information Management & Computer Security*, 22(1), 42–75.
- Stanton, J. M., et al. (2005). Analysis of end-user security behaviors. *Computers & Security*, 24(2), 124–133
- Tsohou, A., et al. (2015). Managing cybersecurity awareness. *Information Management & Computer Security*, 23(3), 273–295.
- U.S. Department of Defense. (2021). *DoD Cyber Strategy*.
- Verizon. 2023 Data Breach Investigations Report. 2023.

APPENDICES

Appendix A: Detailed Research Timeline

Phase	Duration	Weeks	Activity / Task	Milestone / Deliverable
1. Literature Review and Problem Definition	Months 1–2	Weeks 1–8	Conduct comprehensive literature review on human-layer cybersecurity, insider threats, policy compliance, AI/ML hybrid frameworks, and military cybersecurity; identify research gaps	Completion of Chapters 1 & 2; Research problem and objectives defined
2. System Requirements and Framework Design	Month 3	Weeks 9–12	Define functional and non-functional requirements; select high-value human-layer controls; design conceptual and system architecture of hybrid framework	Approved system architecture and conceptual framework diagrams
3. Testbed Development and Data Preparation	Months 4–5	Weeks 13–20	Set up laboratory-based military-style network environment; configure logging, telemetry, and data collection; generate synthetic datasets and scripted behavioural scenarios	Operational testbed with sample datasets ready for experimentation
4. Framework Implementation	Months 6–7	Weeks 21–28	Implement rule-based policy enforcement module; implement machine learning module for anomaly detection and risk scoring; integrate both components into hybrid framework	Functional hybrid detection prototype
5. Experimental Evaluation and Data Analysis	Month 8	Weeks 29–32	Conduct controlled experiments using synthetic and scripted scenarios; analyse results using statistical and ML evaluation metrics	Experimental results report with performance metrics (precision, recall, false positives, latency)

6. Dissertation Writing and Final Revisions	Months 9–10	Weeks 33–40	Draft dissertation chapters (1–8); incorporate supervisor feedback; final proofreading and formatting; prepare submission	Submission-ready dissertation
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Research Timeline (Gantt Chart)

	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Month 7	Month 8	Month 9	Month 10
Phase 1: Literature Review & Problem Definition										
Phase 2: System Design & Requirements										
Phase 3: Testbed Development										
Phase 4: Framework Implementation										
Phase 5: Experimental Evaluation										
Phase 6: Documentation & Writing										

Appendix B: Synthetic Dataset Specification

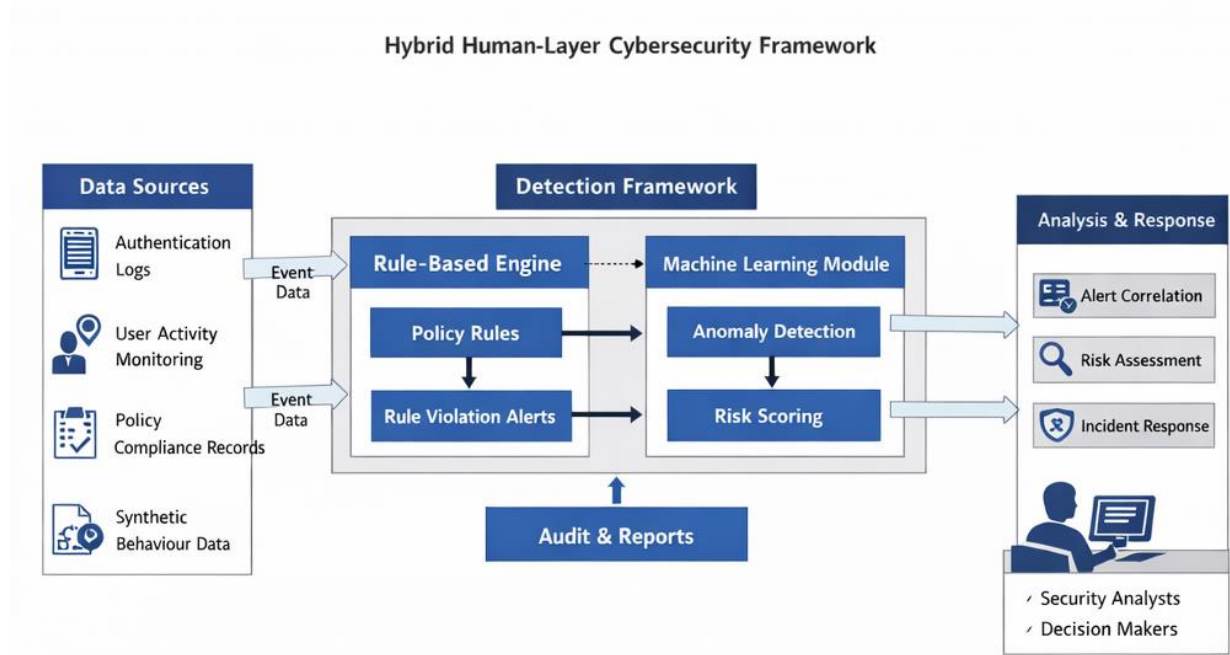
Section	Description / Details
Purpose	Support design, implementation, and empirical evaluation of the hybrid framework using measurable, system-generated evidence.
Rationale	- Avoids classified or sensitive operational data - Provides repeatable evaluation - Supports both rule-based and ML assessment
Dataset Generation	- Controlled lab testbed + synthetic data generation - Scripted behaviour scenarios - Policy-driven violation events

	- Randomised variation for realism
User Roles	- Standard User: Non-privileged staff - Privileged User: Admin/technical staff - Command-Level User: Elevated access - Training-Incomplete User: Missing policy attestations
Event / Log Types	1. Authentication: login attempts, timestamps, success/failure 2. Privilege & Access: admin access, escalation events 3. Endpoint Interaction: USB, software execution 4. Policy Compliance: training, attestations
Policy Violation Scenarios	- Excessive failed logins - Anomalous login behaviour - Excessive/unusual privilege use - Unauthorised endpoint actions - Incomplete training or attestations
Dataset Fields	Event ID, User Role ID, Event Type, Timestamp, Source System, Action, Policy Control Reference, Violation Indicator
Dataset Volume	- Total events: 50,000–100,000 - Normal events: 85–90% - Policy violations: 10–15% - Temporal variation included
Framework Usage	- Rule-based validation - ML feature extraction - Hybrid detection evaluation - Measurement of precision, recall, false positives
Ethical / Security	- Fully synthetic - No PII - Complies with ethics guidelines
Limitations	- Simulated behaviour only - Does not capture live operational complexity - For framework validation only

Appendix C: Human-Layer Policy Control Set

Control ID	Control Name	Control Objective	Observable Behaviour / System Signal	Primary Telemetry Source	Detection Method
HLP-01	Credential Use Compliance	Ensure credentials are used securely	- Repeated failed logins - Logins outside hours - Concurrent logins	Authentication logs (AD/LDAP, VPN)	Rule-based + ML anomaly scoring
HLP-02	Privileged Access Control	Prevent misuse of administrative privileges	- Abnormal admin access frequency - Privilege escalation	Privilege logs, audit logs	Rule-based + ML risk scoring
HLP-03	Endpoint Interaction Control	Enforce safe use of endpoints and peripherals	- Unauthorised USB usage- Execution of prohibited software	Endpoint security logs	Rule-based detection
HLP-04	Mandatory Policy Compliance	Ensure completion of required controls	- Missing training- Incomplete attestations	Training records, compliance systems	Rule-based detection
HLP-05	Behavioural Risk Indicator (Optional)	Detect elevated-risk behaviour	- Minor anomalies across multiple sessions	Aggregated event logs	ML-based risk scoring

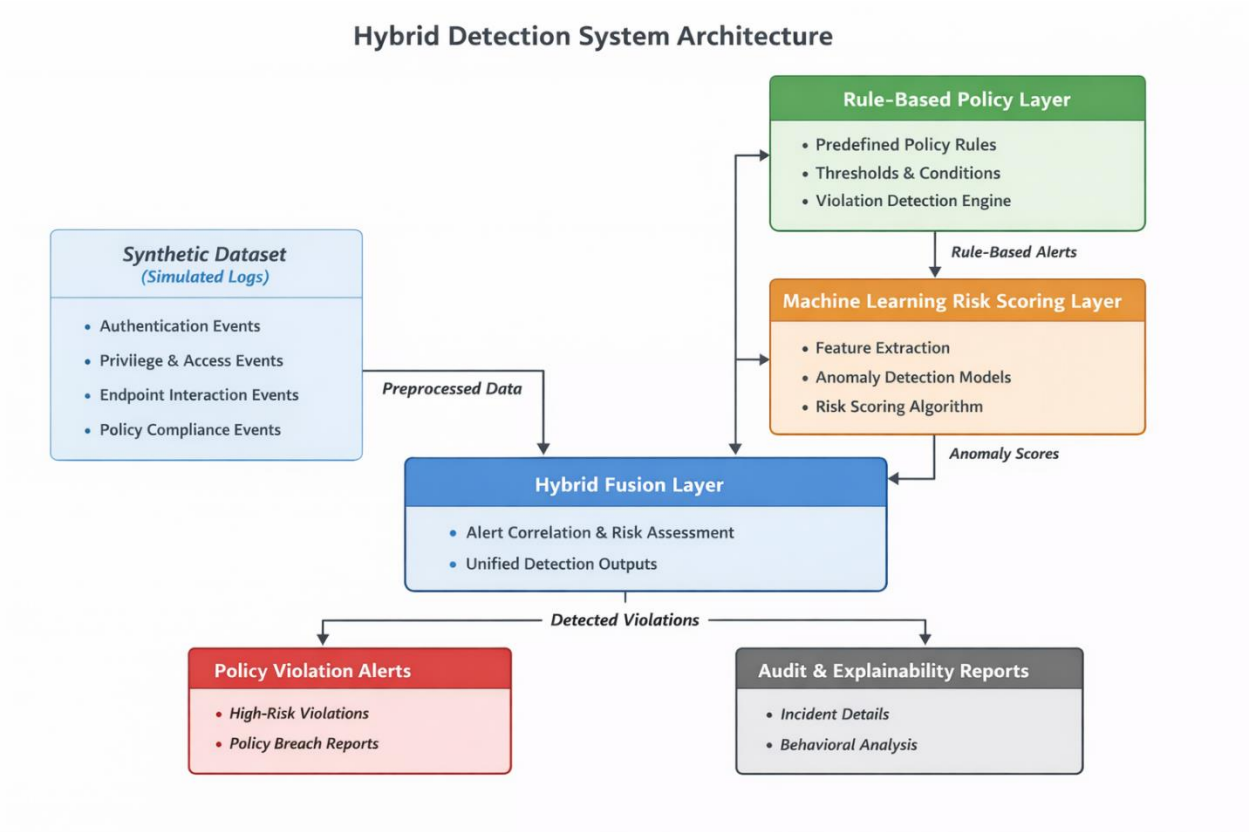
Appendix D: Human-Layer Cybersecurity Framework



Appendix E: Control-to-Research Question Mapping Table

Control ID	Control Name	Linked Research Question(s)	Mapped Objective(s)	Framework Component
HLP-01	Credential Use Compliance	RQ1: Detection of credential misuse	SO1: Evaluate credential violation detection	Rule-Based / ML Layer
HLP-02	Privileged Access Control	RQ2: Detection of privilege misuse	SO2: Evaluate privilege violation detection	Rule-Based / ML Layer
HLP-03	Endpoint Interaction Control	RQ3: Detection of unsafe endpoint actions	SO3: Evaluate endpoint violation detection	Rule-Based Layer
HLP-04	Mandatory Policy Compliance	RQ4: Detection of compliance gaps	SO4: Evaluate compliance violation detection	Rule-Based Layer
HLP-05	Behavioural Risk Indicator	RQ5: Identification of elevated-risk behaviour	SO5: Assess ML-based risk scoring	ML / Hybrid Detection Layer

Appendix F: System Architecture Diagram



Appendix G: Conceptual Framework

Conceptual Framework for Hybrid Human-Layer Cybersecurity Detection

